Cryogenics 104 (2019) 102983

Contents lists available at ScienceDirect

Cryogenics

journal homepage: www.elsevier.com/locate/cryogenics

Technical note

Development of a space 100 mW@5 K closed loop JT cooler

Ma Yuexue*, Quan Jia, Wang Juan, Liu Yanjie, Li Jianguo, Liang Jingtao

Key Laboratory of Space Energy Conversion Technologies, Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, Beijing 100190, China

ARTICLE INFO

Keywords:
4.5 K
100 mW
Multi-orifice
Space

ABSTRACT

A space 100 mW@4.5 K JT cooler is designed, manufactured and experimentally tested by our laboratory. The multi-orifice JT element is proposed to improve the performance of the JT loop. And compared to single orifice, the multi-orifice JT elements perform better. Cooling power of 100.3 mW@4.55 K is achieved when JT element is 4 * 20 μm orifice and supply pressure is 2.0 MPa in open loop experiment. In closed loop experiment, three oil-free linear compressors are used to drive the JT loop, and cooling capacity 102.3 mW at 5.02 K is achieved. This JT cooler will be the potential cryocooler for the future space detectors working at 4.5 K.

1. Introduction

Cryogenics and its applications in space have made a remarkable amount of progresses over the last decades. In the area of space astronomy, some electronic detectors should be cooled to liquid helium temperature level to improve their sensitivity and reduce the background noise [1], so miniature liquid helium temperature cryocooler for space application has become a key field for cryogenic research at National Aeronautics and Space Administration (NASA), Japan Aerospace Exploration Agency (JAXA) and European Space Agency (ESA). In view of its flexibility and relatively higher efficiency, hybrid JT cryocooler is widely used in space detectors working at liquid helium temperature. In fact, nearly all the space applications of mechanical cryocoolers working at liquid helium temperature, having been launched or under development are hybrid JT cryocoolers. For instance, mechanical cryocooler used in Planck is a ^{4}He JT cooler precooled by adsorption cryocooler [2]. JWST uses a three-stage pulse tube cryocooler to precool JT loop [3]. In Superconducting Sub-millimeter-wave Limb-Emission Sounder (SMILES) and Space Infrared Telescope for Cosmology and Astrophysics (SPICA), two-stage Stirling coolers are used to provide precooling power for JT coolers [4–7].

More cooling power means more extensive application prospects, so cryocooler which can provide larger cooling capacity is preferred. At present, the maximum cooling capacity of the reported space hybrid JT cooler is 50.1 mW@4.5 K [6]. However, there is an increasing demand for hundred-milliwatt class cryocooler at 4.5 K, such as THz detectors. Besides, the 4 K JT cooler can take the place of liquid helium dewar to precool space adiabatic demagnetization refrigerator (ADR) or < 1 K adsorption refrigerator.

In China, the Key Laboratory of Space Energy Conversion Technologies (SECT) of the Technical Institute of Physics and Chemistry (TIPC), Chinese Academy of Sciences (CAS) has previously developed a hybrid 4.5 K JT cryocooler [8]. Based on this cooler, a 100 mW@4.5 K JT cooler is designed, manufactured and experimentally tested in this paper.

2. The hybrid JT cooler

2.1. System description

The hybrid JT cooler developed by our laboratory is illustrated as Fig. 1. It is composed of two thermally coupled parts: the multistage pulse tube cryocooler working as precooler and the JT loop. The JT loop is driven by oil-free linear compressors, which makes it suitable for space application. High pressure gas passes through the inner tube of the three counter-flow heat exchangers where it is cooled by the low pressure returning gas. And two additional heat exchangers mounted on the cold end of the precooler provide precooling for JT cycle. When helium gas reaches JT orifice, it has been cooled to nearly 5 K. Then the high pressure helium gas goes through the JT orifice where it will be converted into saturated helium by JT effect.

2.2. Precooler

The multistage high frequency pulse tube cooler developed by our laboratory will be used to precool the JT loop [9]. But in this study, special attention is paid to the development of the JT loop, so a two-stage GM cryocooler is used to precool JT loop in our initial development. If sufficient precooling power can be provided, which kind of precooler makes no difference for JT loop. Besides, the GM cryocooler is

* Corresponding author.
E-mail address: mayuexue@mail.ipc.ac.cn (Y. Ma).

https://doi.org/10.1016/j.cryogenics.2019.102983
Received 25 July 2019; Received in revised form 23 September 2019; Accepted 27 September 2019
Available online 27 September 2019
0011-2275/ © 2019 Elsevier Ltd. All rights reserved.

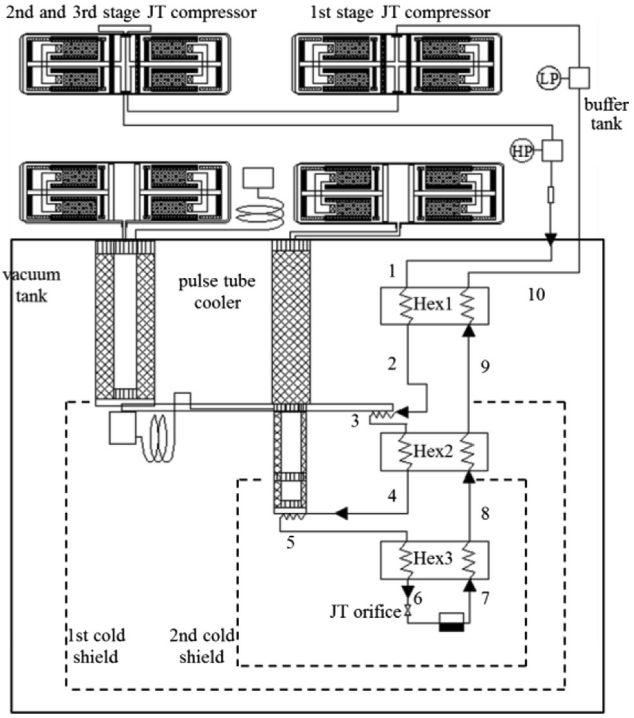
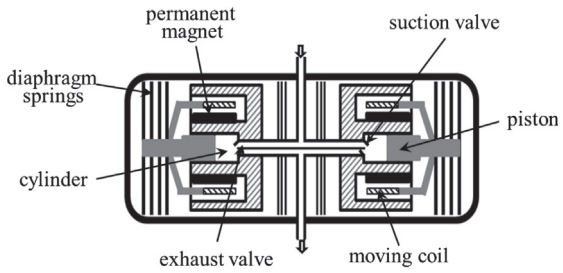


Fig. 1. Schematic of the hybrid JT cooler.

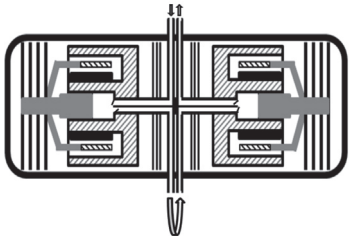
able to shorten the cooling down time and provide sufficient cooling power for JT loop as mass flow rate of JT cooler increases. As mass flow rate of JT cycle changes, the two stages of precooling temperature can be maintained constant by adjusting the electrical heaters attached at the cold ends of GM cooler.

2.3. Space JT compressors

The development of the JT compressor suitable for space application is one of the main challenges in the development of the space 4.5 K JT cooler. Based on the JT compressors in the previous system [8], a new JT compressor system has been developed by our laboratory, as shown in Fig. 2. It consists of two units which connected in tandem. It



a) First stage JT compressor (CP1)



a) Second stage (CP2) and third stage (CP3) JT compressor

Fig. 2. Schematic of the JT compressors.

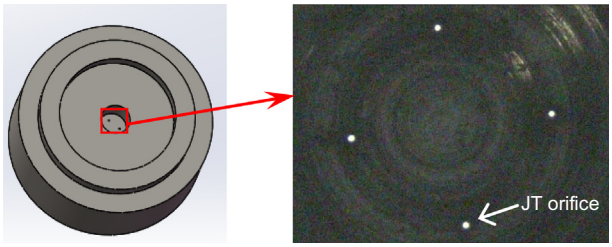


Fig. 3. Schematic diagram and photograph of the four-orifice JT element.

provides three-stage compression for the helium gas. The design value of the suction pressure the first stage JT compressor (CP1) is 0.1 MPa and the third stage JT compressor (CP3) provides a high pressure of 2.0 MPa. All the operating frequency of the three JT compressors is 50 Hz. The diameters of the pistons are 9 mm, 19 mm and 26 mm respectively. The pistons are supported by diaphragm springs. So the JT compressor is completely oil free by use of clearance seal between the piston and cylinder. This three-stage JT compressor system is used to drive the JT loop in the following closed loop experimental study.

2.4. JT orifice

The JT orifice not only is a restriction that allows the high pressure gas to expand to low pressure gas but also a decisive factor which affects the cooling capacity of the JT cooler. According to our previous study, cooling power of 100 mW may not be achieved by simply increasing diameter of the JT orifice [10]. Although mass flow rate increases as the diameter of the JT orifice increases, the JT effect will deteriorate. Thus, JT element with multi-orifice is proposed to improve the cooling capacity of the JT loop, as shown in Fig. 3. Several smaller orifices instead of one larger orifice are used to provide the pressure drop for the JT cooler, as shown in Fig. 3.

2.5. Heat exchangers

The counter-flow heat exchangers are coaxial tube-in- tube design, with high pressure helium gas flowing in the annular space between the two tubes and the low pressure helium gas flowing in the inner tube. In this study, we make sure that the efficiency of the counter-flow heat exchangers will not be lower than 0.97 as the mass flow rate changes. And the design of the third exchanger (Hex3) is quite important because its efficiency influences cooling power of the JT cooler. Besides, the JT loop is thermally coupled with the precooler through two heat exchangers attached on the cold ends of the precooler.

2.6. Design value of the JT cooler

To achieve cooling power of 100 mW, the second stage precooling temperature is assumed to be 15 K considering the limitation of the precooler and the efficiency of the whole hybrid JT cooler. The supply pressure is assumed to be 2.0 MPa considering the conversion pressure of helium and the limitation of our JT compressors. Then, the mass flow rate of the JT loop and the diameter of the JT orifice are calculated to be 10.5 mg/s and 34 μm respectively. The detail design value of the JT cooler is given in Table 1.

3. Experimental study

For the purpose of achieving 100 mW at 4.5 K, both the 34 μm JT orifice and the multi-orifice JT element are experimentally tested. The diameter of the JT orifice has an error range of ± 2 μm. Firstly, open loop experiment is carried out because it's more convenient to adjust the supply pressure precisely. Then, closed JT loop driven by three stage linear compressors are experimentally studied.

Table 1
Design value of the JT cooler.

Mass flow rate	10.5 mg/s	
Precooling temperature	First stage	Second stage
	100 K	15 K
Precooling power	First stage	Second stage
	350 mW	260 mW
Pressure ratio	2.0 MPa/0.13 MPa	
Diameter of the JT orifice	34 μm	
Cooling capacity	100 mW@4.5 K	

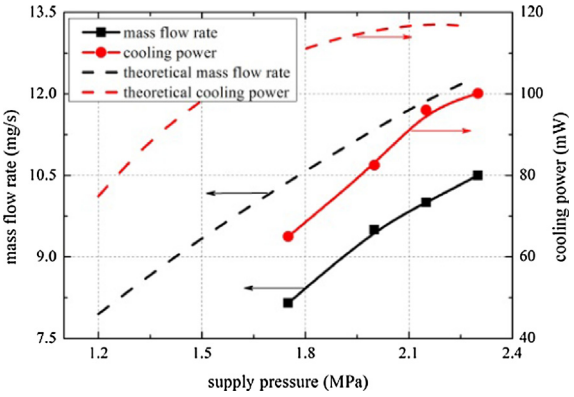


Fig. 4. Influence of supply pressure on cooling power of the JT cooler.

3.1. Open loop experiments

Firstly, open loop experiment is carried out because it's more convenient to adjust the supply pressure precisely. A helium gas cylinder instead of JT compressor is used to provide high pressure gas and the low pressure gas vents directly to atmosphere. Fig. 4 gives the theoretical values and experimental results of 34 μm JT element. The theoretical mass flow rate is relatively higher due to the ignorance of the flow resistance of return gas and the compressibility of the gas. Thus, theoretical cooling power is also higher than experimental results. The mass flow rate and cooling power rise with increasing supply pressure. But cooling capacity of only 82mW is achieved when supply pressure is 2.0 MPa which is attributed to the fact that the mass flow rate is lower than theoretical value. Thus, higher supply pressure is provided for the JT loop. Then, cooling capacity of 100.1 mW@4.51 K is achieved when supply pressure is 2.3 MPa. However, supply pressure of 2.3 MPa is a great challenge for the space JT compressors. Therefore, bigger JT orifice is used to increase mass flow rate of the JT cooler when supply pressure is 2.0 MPa. Fig. 5 gives the performance of different JT orifices. Although mass flow rate increases as diameter of the JT orifice

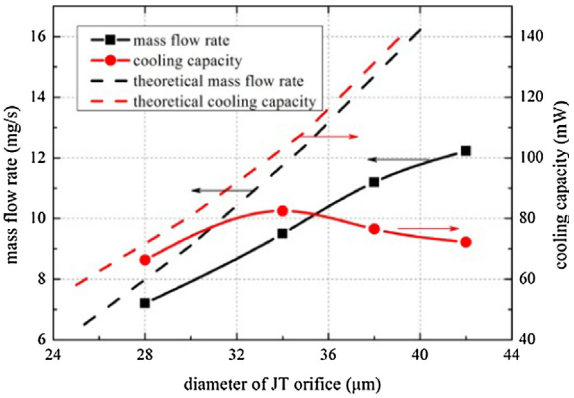


Fig. 5. Influence of diameter of JT orifice on cooling power of the JT cooler [10].

Table 2
Performance comparison of the 34 μm and 4 * 20 μm.

JT element	Mass flow rate	Cooling capacity	Supply pressure
34 μm	9.5 mg/s	82.5 mW @ 4.44 K	2.0 MPa
4 * 20 μm	11.08 mg/s	100.3 mW@ 4.55 K	2.0 MPa

increases, cooling power will not increases monotonously as theoretical values predict. The JT effect will deteriorate because the flow resistance of the JT orifice drops as its diameter increases. Cooling capacity decreases when the JT orifice is larger than about 35 μm.

Then, 4 * 20 μm JT element is experimentally tested. Table 2 gives the performance of the four-orifice JT element and single orifice. Although the cross section area the 4 * 20 μm is larger than 34 μm orifice, the cooling power of the JT cooler doesn't decrease as predicted in Fig. 5. And cooling capacity of 100.3 mW@4.55 K is achieved when supply pressure is 2.0 MPa.

3.2. Closed loop study

The open loop is not suitable for practical applications, so closed loop JT cooler is studied in this part. The three-stage JT compressor system is used to drive the JT loop. The three-stage JT compressor system is used to drive the JT loop. The working parameters of the 34 μm and 4 * 20 μm JT elements are given in Table 3. In closed loop experiments, the performance of the JT cooler is about the same with open loop study. The operating frequency of the three-stage JT compressors is 50 Hz. And the pressure ratios of the three-stage compressors are about 2.5, 3.7 and 1.8, respectively.

Cooling power of 102.3 mW is achieved when 4 * 20 μm JT element is used. But the operating temperature is 5.02 K due to the limitation of our JT compressors. Suction pressure of only 0.16 MPa can be achieved when supply pressure is 2.0 MPa. And the total power consumption of JT compressors is about 146 W.

4. Discussion

All in all, increasing mass flow rate is one of the efficient ways to improve the cooling capacity of the JT cooler. However, although mass flow rate rises with increasing diameter of JT orifice, cooling capacity will decrease when the flow resistance of the JT orifice is lower than a certain value which leads to the deterioration of the JT effect. So, smaller JT orifice which can provide larger pressure drop for the JT loop performs better in the JT cooler. But the mass flow rate will be limited by using smaller JT orifice which will lead to the limitation of the cooling capacity. And multi-orifice JT element can increase the mass flow rate without deteriorating JT effect.

Finally, a 100 mW JT cooler working at liquid helium temperature is developed. Although the operating temperature is 5.02 K due to the limitation of the JT compressors, it is still an attractive value considering the fact that the JT cooler is driven by oil-free compressors which are suitable for future space applications. However, there is still much more work needs to be done to explore the influence mechanism of the multi-orifice. And the influence of the JT orifice on the mass flow rate and the cooling capacity of the JT cooler is quite complicated. It may associate with the increase of viscous dissipation due to the enhanced friction of the multi-orifice.

Besides, the performance of our pulse tube cooler should be improved because the JT loop requires much more precooling power as mass flow rate increases. Therefore, the optimization of the JT compressors and the development of the more efficient and powerful pulse tube are in urgent demand.

Table 3
Working parameters of the 34 μm and 4 * 20 μm JT elements.

JT element	Electrical power	Mass flow rate	P_h/P_l (MPa)	Cooling capacity
34 μm	103 W	9.5 mg/s	2.0/0.12	81.5 mW@4.48 K
4 * 20 μm	146 W	11.02 mg/s	2.0/0.16	102.3 mW @ 5.02 K

5. Conclusion

A 100 mW@4.5 K hybrid JT cryocooler is developed by our laboratory. Detail analysis is carried out on the JT loop for the purpose of improving its cooling power. Although mass flow rate increases monotonously with increasing diameter of JT orifices, cooling capacity will decrease when the size of the JT orifice is larger than a certain value. The multi-orifice JT element is proposed to improve the performance of the JT cooler. And based on experimental research, multi-orifice JT element performs better than single orifice. Finally, JT element of 4 * 20 μm is experimentally tested in detail. Cooling power of 100.3 mW at 4.55 K is achieved in open loop experiment when supply pressure is 2.0 MPa. And cooling capacity of 102.3 mW@5.02 K is achieved when the JT loop is driven by three stage oil-free compressors. The total power consumption of the JT coop is 146 W.

Acknowledgement

This work was supported by National Natural Science Foundation of China (Grant No. 51776213) and the Strategic Priority Research Program of Chinese Academy of Sciences (Grant No. XDA18000000).

References

[1] Ross Jr RG, Boyle RF, Kittel P. NASA Space cryocooler programs –a 2003 overview [J]. Adv Cryogenic Eng 2003;49B(710):1197–204.
[2] Morgante G, Pearson D, Melot F, et al. Cryogenic characterization of the Planck sorption cooler system flight model [J]. J Instrum 2009;4(12). T12016-T12016.
[3] R.G. Ross Jr, A Study of the Use of 6K ACTDP Cryocoolers for the MIRI Instrument on JWST [M], Springer US, 2005.
[4] Otsuka K, Tsunematsu S, Okabayashi A, et al. Test results after refurbish of cryogenic system for smiles [J]. Cryogenics 2010;50(9):512–5.
[5] Shinozaki K, Sugita H, Sato Y, et al. Developments of 1–4 K class space mechanical coolers for new generation satellite missions in JAXA[J]. J Phys Chem B 2008;112(15):4561.
[6] Sugita H, Sato Y, Nakagawa T, et al. Cryogenic system design of the next generation infrared space telescope SPICA [J]. Cryogenics 2010;50(9):566–71.
[7] Sugita H, Nakagawa T, Murakami H, et al. Cryogenic infrared mission "JAXA/SPICA" with advanced cryocoolers [J]. Cryogenics 2006;46(2):149–57.
[8] Quan J, Zhou ZJ, Liu YJ, et al. A miniature liquid helium temperature JT cryocooler for space application [J]. Sci China Technol Sci 2014;57(11):2236–40.
[9] Quan J, Liu Y, Liu D, et al. 4 K high frequency pulse tube cryocooler used for terahertz space application[J]. Chin Sci Bull 2014;59(27):3490–4.
[10] Ma Y, Wang J, Liu Y, et al. Experimental research on the J-T orifice of a space 4.5 K hybrid J-T cooler[J]. Chin Sci Bull 2018;63(18):1839–64. (in Chinese).